

HPC exam project - part 2

A.Y. 2025-26

The following exercise builds upon the previous one concerning the simulation of damped wave motion using OpenMP. In this second version of the exercise, you are required to perform multiple parallel simulations using the MPI library on an HPC system. Each individual simulation will continue to be executed in parallel using OpenMP, as already implemented in the first part.

In this second part, the program must execute three distinct simulations and produce three separate videos. The frames of each video will be saved in three separate folders named **sim1**, **sim2**, and **sim3**.

The first simulation remains unchanged with respect to the initial OpenMP-based version of the simulator. In the second simulation, two damped waves are generated at two different points in space. The two waves share the same damping coefficient (γ) and the same propagation speed (c). Both waves are initiated at time $t = 0$ and propagate across the plane, simulating wave interference phenomena.

The third simulation is similar to the previous one, but the second wave is initiated at time $t = t_start$.

Each student will find their specific simulation parameters in the file **Paramiter_part_2.pdf** to be used in their project.